In this manuscript, the authors implement a score-based generative model (SGM) in a network of nonlinear oscillators coupled to a thermal bath trained via local learning rules that update parameters in an explicit energy function composed of local and coupling terms modeled by low-order polynomials. They show how by adding noise to samples drawn from a target distribution in a forward process, a system described by their energy function can be trained to generate samples from this target distribution by applying a reverse process where a temporal force protocol is applied. This training requires measuring the forces in the system clamped to the forward sample x_t at each time step t. Without access to the forces, the objective gradient can be approximated using (one-step) contrastive divergence. The authors showcase their algorithm with two learning tasks. The first consists of learning a simple two-dimensional bimodal Gaussian distribution; they show that their SGM closely matches the target distribution while an alternative equilibrium sampling (ES) algorithm fails at this task due to the system getting trapped in an energy basin. The second consist of a classic Machine Learning problem: learning to generate MNIST samples of the numbers 0 and 1. The authors train a network of oscillators and show representative generated images in Figure 2.

This paper is well-written, although at times difficult to follow for a non-expert in SGMs. I feel the paper would benefit from additional clarification why certain steps are taken. For example, explain why you chose oscillators as your physical platform; why asymmetric couplings are included in eq. (4). Also, because this paper builds upon existing literature in SGMs, it is not always clear what already existed in this field and what is new; this could be improved simply by adding a few clarifying sentences.

However, my biggest concern with this paper is that no effort is made to impact any field beside those working on theoretical and numerical physical learning. One important motivation of this field is that these learning algorithms must be applicable to physical systems; yet the discussion of experimental realizations merely lists examples and includes no critical evaluation of the algorithm. In my opinion, if experimental implementation is not the goal, then this specific local-rule algorithm either must have a clear benefit over other algorithms numerically (comparing SGM to ES in this system is not sufficient: it does not give any new insights), or it must offer new fundamental insights into learning in physical systems. I am not convinced this paper does either.

Therefore, in its current state I do not think this paper is a good fit for PRL. It is well written, and the proposed algorithm holds potential for learning in physical systems, but the authors have not sufficiently shown why this algorithm is particularly relevant, insightful or efficient. Without significant extra work I do not see this changing. Instead, it is my opinion that this manuscript is better served in a more specialized journal such as PRE. Please find below some small comments.

Comments:

1. In Fig. 1 (c) and (f), the iso-contour appears to be yellow, not red as described in the caption. The yellow lines around the two minima also do not help at all.

Moreover, the Boltzmann distribution (colorbar) appears to be uniform (all blue).

2. A quantitative measure that shows how good your generated 0 and 1 are would be appreciated.
3. How does the computational complexity of your coupled oscillators scale with connectivity, parameters, etcetera.